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New Suspension Design for Heavy Duty Trucks: Dynamic Considerations

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ABSTRACT

It is well known that the excessive levels of vibration in heavy vehicles negatively affect driver comfortability, cargo safety and road condition. The current challenge in the field of suspension design for heavy vehicles is to optimize the suspension dynamic parameters to improve such requirements.

Almost all of the previous work in this field is based on applying the mathematical optimization considering active or passive suspension systems to obtain the optimal dynamic parameters.

In this work a new passive suspension systems for heavy trucks is suggested and compared with the conventional passive suspension systems. The new systems rely on transferring the vertical motion, (vibration), into horizontal motion through a bell-crank mechanism to be taken by a horizontal passive suspension system. The system dynamic parameters like body acceleration, suspension travel and dynamic tire load are calculated assuming random excitation due to road irregularities.

The new suspension system is examined using single wheel station and tandem axle configurations.

Improvements in body acceleration, suspension working space and dynamic tire load have been achieved compared with the conventional passive systems. These improvements are qualitatively almost equal to the improvement that usually achieved by using active suspension system.

The new design is worth applying to cut the cost consumed in employing expensive active suspension systems.

INTRODUCTION

During the past three decades, significant developments in the constructions of the suspension systems have

been apparent. Furthermore maintaining lower components costs was increasingly challenging. For this purpose, the automotive suspension designers pay great attention to optimize the suspension system parameters. That is to increase both ride quality and vehicle safety and to reduce roads wear or damage. In the seventeen's and eighteen's, comprehensive researches were concerned with analyzing vehicle responses excited by the road surface irregularities [1-5]. In these papers different techniques were developed to provide the trucking industry with tools to predict the vehicle dynamic behavior.

Several optimization studies, considering passive suspension systems were made for optimizing suspension system parameters [6-10]. Different Objective functions were used and optimal suspension parameters, particularly damping coefficients, were generated. Using the optimized suspension parameters, the vehicle ride quality was improved. Unfortunately, due to the limitations of the passive suspension systems, it is difficult to tune it while it interact with vehicle structure covering wide excitation frequency ranges. The active suspension systems were designed for solving such vibration problem [11-14]. Since the active suspension systems are too expensive, suspension designers used to design intelligent controllable dampers. The requirements for offering controllable dampers, sensors and control units still cause high costs.

In this paper, new suspension systems have been designed for heavy trucks. The new systems transform most the vertical motions into horizontal motions through a bell crank mechanism. The ride behavior of the new suspension systems is examined using quarter car models considering single wheel station and tandem axle configurations. Using different geometry and isolation parameters (stiffness and damping coefficients), significant improvements in the ride quality are obtained. The random responses of the present

vehicle models are concerned with the following variables:

- Body acceleration,
- Suspension working spaces,
- Dynamic tire loads.

CONVENTIONAL SUSPENSION SYSTEMS

This section describes two mathematical models for conventional suspension systems; (a) single wheel station suspension system and (b) linear tandem bogie system (tandem suspension system). The ride behavior of these models is compared with that of the new suspension systems.

CONVENTIONAL SINGLE WHEEL STATION SUSPENSION SYSTEM

The model is 2 degrees of freedom system, Figure 1 [15]. The degrees of freedom represent the truck body motion, z_s , and the motion of wheel/axle assembly, z_u . The vehicle is suspended on the wheel/axle assembly through a vertical spring fitted in parallel with a viscous damper. The tire is modeled as a spring and damper.

The formulation of the equations of motion and the response variables of the conventional single wheel station system can be found in [15]. The model parameters represent the rear of a two axles truck and given in Table 1 [16].

Table 1: Parameters of the conventional single wheel station suspension system [16]

Body mass,	5979 kg
Mass of the wheel/axle assembly,	596 kg
Suspension damping ,	29 kNs/m
Suspension stiffness,	0.583 MN/m
Tire stiffness,	2.4 MN/m
Tire damping,	1.2 kNs/m

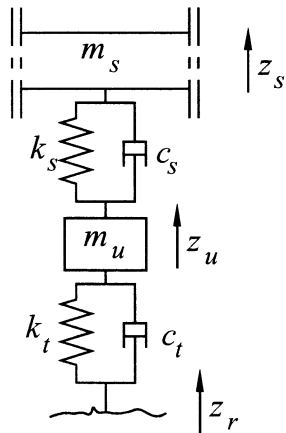


Fig. 1 Conventional single wheel station suspension system [15].

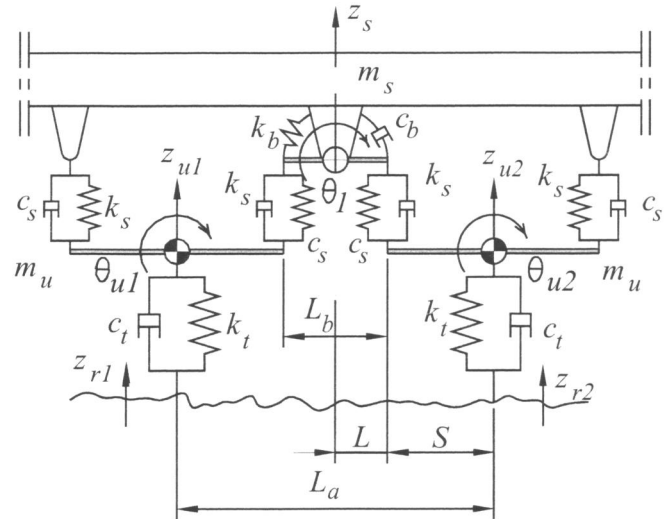


Fig. 2 Conventional tandem suspension system [16]

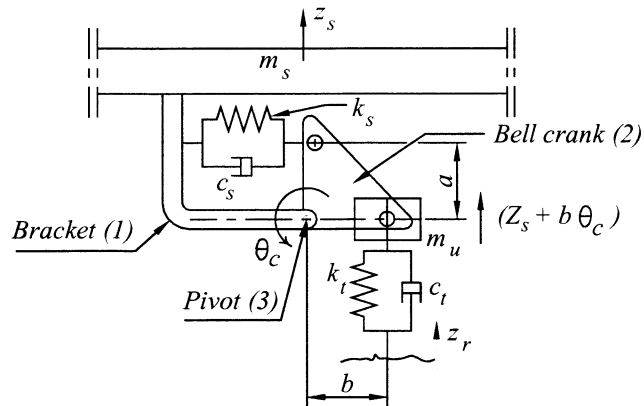


Fig. 3 New single wheel station suspension system.

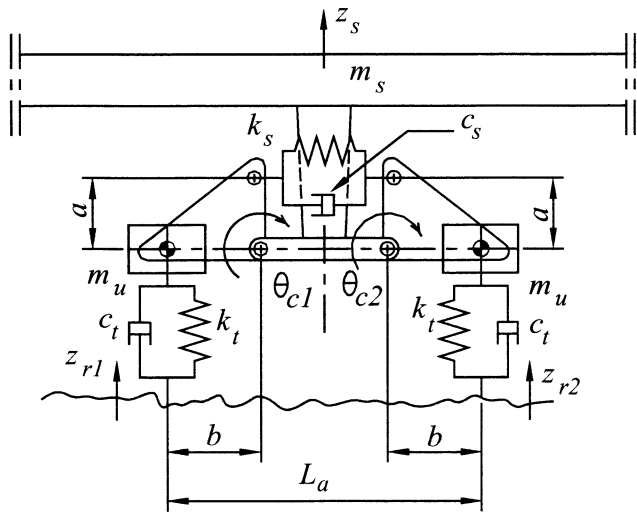


Fig. 4 New tandem suspension system.

CONVENTIONAL TANDEM SUSPENSION SYSTEM

An air suspended tandem axle is used to generate a ride vibration model for a vehicle equipped with two axle unit. The model is 6 degrees of freedom linear tandem bogie system, Figure 2 [17]. The used model parameters are given in Table 2. The degrees of freedom represent the vertical motions of the vehicle body and wheel/axle assemblies and the angular motion of the suspension beams and the balancing beam.

The equations of motion of the vehicle system are generated using the Lagrange's energy equation which is expressed, for linear systems, as

$$\frac{d}{dt} \left(\frac{\partial KE}{\partial \dot{z}_j} \right) + \frac{\partial DF}{\partial \dot{z}_j} + \frac{\partial PE}{\partial z_j} = F_j$$

where

KE , PE are kinetic and potential energies, DF is the dissipative function, z_j is the generalized coordinate number j , F_j is the generalized force acting on the generalized coordinate number j .

Since the systems in this work are passive systems, there are no generalized forces. The excitation is generated due the random road surface irregularities. The system energies are evaluated as follows:

Kinetic energy:

$$KE = \frac{1}{2} m_s \dot{z}_s^2 + \frac{1}{2} I_b \dot{\theta}_1^2 + \frac{1}{2} m_u \dot{z}_{u1}^2 + \frac{1}{2} I_u \dot{\theta}_{u1}^2 + \frac{1}{2} m_u \dot{z}_{u2}^2 + \frac{1}{2} I_u \dot{\theta}_{u2}^2$$

Potential Energy:

$$PE = \frac{1}{2} k_b \theta_1^2 + \frac{1}{2} k_t (z_{u1} - z_{r1})^2 + \frac{1}{2} k_t (z_{u2} - z_{r2})^2 + \frac{1}{2} k_s [z_s - (z_{u1} + S\theta_{u1})]^2 + \frac{1}{2} k_s [z_s - (z_{u2} - S\theta_{u2})]^2 + \frac{1}{2} k_s [(z_s + L\theta_1) - (z_{u1} - S\theta_{u1})]^2 + \frac{1}{2} k_s [(z_s - L\theta_1) - (z_{u2} + S\theta_{u2})]^2$$

Dissipative Function:

$$DF = \frac{1}{2} c_b \dot{\theta}_1^2 + \frac{1}{2} c_t (\dot{z}_{u1} - \dot{z}_{r1})^2 + \frac{1}{2} c_t (\dot{z}_{u2} - \dot{z}_{r2})^2 + \frac{1}{2} c_s [\dot{z}_s - (\dot{z}_{u1} + S\dot{\theta}_{u1})]^2 + \frac{1}{2} c_s [\dot{z}_s - (\dot{z}_{u2} - S\dot{\theta}_{u2})]^2 + \frac{1}{2} c_s [(\dot{z}_s + L\dot{\theta}_1) - (\dot{z}_{u1} - S\dot{\theta}_{u1})]^2$$

$$+ \frac{1}{2} c_s [(\dot{z}_s - L\dot{\theta}_1) - (\dot{z}_{u2} + S\dot{\theta}_{u2})]^2$$

Substituting the partial derivative of the above energy equation in the Lagrange's equation, the equations of motion can be generated.

The response variables are

$$\text{Body vertical acceleration} = -\omega^2 z_s$$

$$\text{Front suspension working space} = z_s - z_{u1}$$

$$\text{Rear suspension working space} = z_s - z_{u2}$$

$$\text{Front dynamic tire force} = k_t (z_{u1} - z_{r1}) + c_t (\dot{z}_{u1} - \dot{z}_{r1})$$

$$\text{Rear dynamic tire force} = k_t (z_{u2} - z_{r2}) + c_t (\dot{z}_{u2} - \dot{z}_{r2})$$

Table 2: Parameters of the conventional tandem suspension system [17]

Body mass,	8200 kg
Mass of the wheel/axle assembly,	400 kg
Mass moment of inertia of the balancing arm,	1 kgm ²
Suspension damping ,	5 kNs/m
Suspension stiffness,	0.4 MN/m
Angular damping of the balancing arm,	100 Nms/rad
Angular stiffness of the balancing arm,	1 Nms/rad
Tire stiffness,	2 MN/m
Tire damping,	1.2 kNs/m
Length of the balancing arm,	0.4 m
Distance between the two axles,	1.4 m

NEW SUSPENSION SYSTEMS

NEW SINGLE WHEEL STATION SUSPENSION SYSTEMS

The configuration of the new suspension system is shown in Figure 3. The vehicle main suspension elements are fitted horizontally between the bracket 1, which is assumed to be welded with the vehicle frame, and the vertical face of the rotary link (bell crank) 2. The center of the angular oscillation of the link 2 is the pivot 3 fixed in the bracket 1.

The wheel mass (m_u) is to be positioned at the wheel axle position. The system degrees of freedom are the vertical motion of the vehicle body and the angular motion of the bell crank 2. Based on this configuration, the suspension system will stores and dissipates energies in the horizontal direction. Also, it is expected that the offset distances (a and b) will play an important role in transmitting vibrations from the wheel to the body or vise versa. The nominal values of the model parameters are given in Table 1. Additional values related to the bell crank and the modified suspension parameters are given in the result section of the present work.

The system energies are expressed as follows:

Kinetic energy:

$$KE = \frac{1}{2} m_s \dot{z}_s^2 + \frac{1}{2} I_c \dot{\theta}_c^2 + \frac{1}{2} m_u (\dot{z}_s + b \dot{\theta}_c)^2$$

Potential Energy:

$$PE = \frac{1}{2} k_t [z_r - (z_s + b \theta_c)]^2 + \frac{1}{2} k_s (a \theta_c)^2$$

Dissipative function:

$$DF = \frac{1}{2} c_t [\dot{z}_r - (\dot{z}_s + b \dot{\theta}_c)]^2 + \frac{1}{2} c_s (a \dot{\theta}_c)^2.$$

The response variables are:

$$\text{Body vertical acceleration} = -\omega^2 z_s$$

$$\text{Suspension working space} = a \theta_c$$

Dynamic tire force =

$$k_t [z_r - (z_s + b \theta_c)] + c_t [\dot{z}_r - (\dot{z}_s + b \dot{\theta}_c)]$$

NEW TANDEM SUSPENSION SYSTEMS

In the proposed new suspension design, the axles of the tandem suspension system are hinged to two opposite bell cranks as shown diagrammatically in Figure 4. The bell cranks are hinged in a bracket welded to the vehicle frame. The top ends of the opposite bell cranks are connected to a horizontal spring/damper flexible joint. The nominal values of the model parameters are given in Table 2. Additional values related to the bell cranks and modified suspension parameters are given in the result section of the present work. The model is three degrees of freedom system. These degrees of freedom are vertical motion of vehicle body and two angular motions of the two bell cranks.

The system energies are expressed as follows:

Kinetic energy:

$$KE = \frac{1}{2} m_s \dot{z}_s^2 + \frac{1}{2} I_c \dot{\theta}_{c1}^2 + \frac{1}{2} m_u (\dot{z}_s + b \dot{\theta}_{c1})^2 + \frac{1}{2} I_c \dot{\theta}_{c2}^2 + \frac{1}{2} m_u (\dot{z}_s + b \dot{\theta}_{c2})^2$$

Potential energy:

$$PE = \frac{1}{2} k_t [z_{r1} - (z_s + b \theta_{c1})]^2 + \frac{1}{2} k_t [z_{r2} - (z_s + b \theta_{c2})]^2 + k_s (a \theta_{c1} - a \theta_{c2})^2$$

Dissipative function:

$$DF = \frac{1}{2} c_t [\dot{z}_{r1} - (\dot{z}_s + b \dot{\theta}_{c1})]^2 + \frac{1}{2} c_t [\dot{z}_{r2} - (\dot{z}_s + b \dot{\theta}_{c2})]^2 + c_s (a \dot{\theta}_{c1} - a \dot{\theta}_{c2})^2.$$

The response variables are:

$$\text{Body vertical acceleration} = -\omega^2 z_s$$

$$\text{Suspension working space} = a \theta_{c1} - a \theta_{c2}$$

Front dynamic tire force =

$$k_t [z_{r1} - (z_s + b \theta_{c1})] + c_t [\dot{z}_{r1} - (\dot{z}_s + b \dot{\theta}_{c1})]$$

Rear dynamic tire force =

$$k_t [z_{r2} - (z_s + b \theta_{c2})] + c_t [\dot{z}_{r2} - (\dot{z}_s + b \dot{\theta}_{c2})]$$

DISCRIPTION OF THE RANDOM ROAD SURFACE

The vehicle is assumed to move over a random road surface. The power spectral density of the road irregularities, considering a single wheel station model, can be expressed as given in [15];

$$PSD_{RS} = G \frac{v^{p-1}}{f^p}$$

where

G is the roughness coefficient, 5×10^{-1} for a principle road,

v vehicle velocity, m/sec,

f frequency, Hz,

p is the slop of the log-log spectral density curve, 2.5.

For the two axles models, the power spectral density of the road excitation signals, including the cross correlation, can be expressed as;

$$PSD_{RD} = PSD_{RS} \begin{bmatrix} 1 & e^{-i\omega D} \\ e^{i\omega D} & 1 \end{bmatrix}$$

where

$$\omega = 2\pi f$$

D is the time delay between the two axles.

RANDOM RESPONSES

The equations of motion of the vehicle system can be written in a matrix form as follows:

$$[M]\{\ddot{z}\} + [C]\{\dot{z}\} + [K]\{z\} = [C_{EX}]\{\dot{z}_O\} + [K_{EX}]\{z_O\}$$

The vector of the power spectral densities of the output variables can be expressed as:

$$PSD_{RES} = [H_T(if)] PSD_{ROAD} [H_T(-if)]^T$$

$[H_T(if)]$ is the transformed transfer function of the vehicle system.

$$H_T(if) = [-(2\pi f)^2 [T_{ACC}] + i(2\pi f) [T_{VEL}] + T_{DIS}] [H(if)] + [T_{EDIS} + i(2\pi f) T_{EVEL}]$$

$$H(if) = [[K] - (2\pi f)^2[M] + i(2\pi f)[C]]^{-1} [[K_{EX} + i(2\pi f)[C_{EX}]]$$

T_{ACC} , T_{VEL} and T_{DIS} are the transformation matrices of the system acceleration, velocity and displacement vectors,

T_{EVE} , T_{EDIS} are the transformation matrices of the excitation displacement and velocity.

The root mean square (*RMS*) values of the response variables can be expressed as

$$RMS_j = \sqrt{\int_{f1}^{f2} PSD_{RES}^j(if)df}$$

where j is the number of the response variable,

$f1$, $f2$ are the minimum and maximum values of the excitation frequency, Hz.

RESULTS

In the following section, the attained results for the new suspension design will be presented and discussed. The main goal of employing such new design is to optimize the dynamic tire load, vertical body acceleration and suspension working space without the need for active suspension system.

NEW SINGLE WHEEL STATION SYSTEM

In Fig 5 a, b and c, the power spectral density of the vertical body acceleration, dynamic tire load and suspension working space are plotted versus frequency for three road types. It is found that using a leverage ratio of the bell crank mechanism, $a/b=1$ gives the same performance like the traditional passive suspension system. From these figures, it can be also noticed that the tested parameters are higher for minor road, $G=5 \times 10^{-6}$ which confirms with the previous findings [15].

The current study focusing on optimizing the passive suspension system to the maximum extent by using different leverage ratios for the bell crank mechanism and using different values for the spring stiffness and damping coefficients to achieve the most minimum values of the parameter under study. This will lead at the end to a very good suspension with minimal cost compared with the active suspension systems.

The first step in optimization technique is to get the leverage ratio that gives minimum levels of dynamic tire load, body vertical vibration and minimum suspension working space. Different values for the leverage ratio, a/b have been used while keeping other system parameters unchanged. Based on results in Figures 6 a,b and c, the best leverage ratio of the bell crank mechanism to give minimum levels of dynamic tyre load and vertical body acceleration is $a/b= 0.25/0.35$. It is important to note that there is a slight increase in the

suspension working space at this ratio that can be acceptable in certain circumstances if compared in the gain in other parameters.

The second step in the optimization technique is to guess the optimum spring stiffness that achieves low levels of the three dominating performance characteristics under study. After trying a quite large number of spring stiffnesses while keeping the rest of the parameter unchanged, it is found that employing very stiff springs deteriorates the dynamic stability of the suspension while relatively softer ones improve it, Fig 7. A stiffness value of about 75% of the nominal value is found to give the lowest level of vertical vibrations and dynamic tire load. No significant improvement in the suspension working space is noticed., Fig. 7 b.

The last step in the used optimization procedure is to optimize the damping coefficient of the used damper. Similar technique is used and it is found that big gain can be achieved by using dampers with relatively higher damping coefficient. Doubling the nominal value of the damping coefficient gives the lowest dynamic tire load, vertical body acceleration and suspension working space, Fig. 8 a, b and c. Low and very high damping coefficients are found to be disadvantageous in the optimization process.

Tables 3, 4 and 5 gives the RMS values of the new system response as being affected by the leverage ratio, spring stiffness and damping coefficient respectively. From which it is clear that there are significant improvement of the dynamic tire load and the vertical body acceleration.

Table 3 Effect of a/b ratio on the RMS values of the response variables

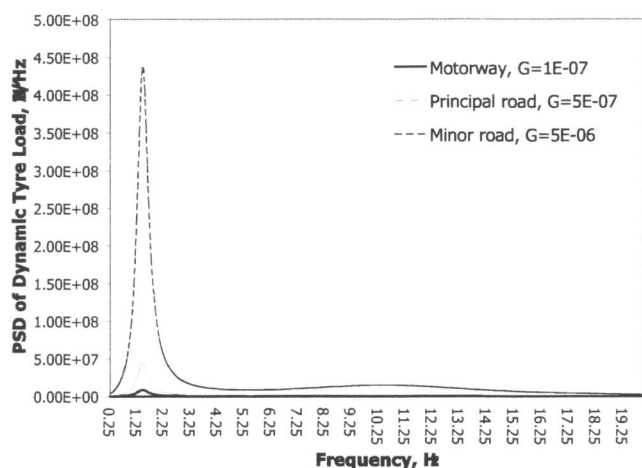
	$a=0.3m$ $b=0.3m$	$= 0.25m$ $= 0.3m$	$= 0.25m$ $=0.35m$	$=0.35m$ $=0.3m$
$\dot{z}_s$	1.0457	0.8831	0.7682	1.2086
<i>sws</i>	0.0087	0.009	0.0093	0.0085
<i>dtl</i>	6835.5	6361.7	6355	7632

Table 4 Effect of spring stiffness the RMS values of the response variables

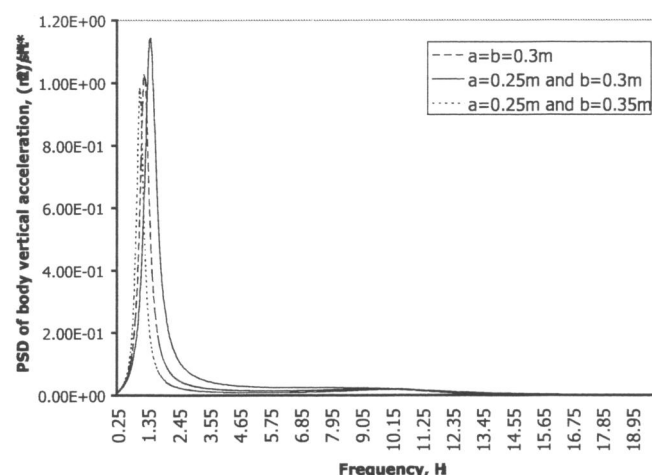
	$k=0.75 k_n$	$k_n = k$	$k=1.25 k_n$
$\dot{z}_s$	0.6948	0.7555	0.8238
<i>sws</i>	0.0066	0.0065	0.0064
<i>dtl</i>	5296.2	5500	5770

Table 5 Effect of suspension damping on the RMS values of the response variables.

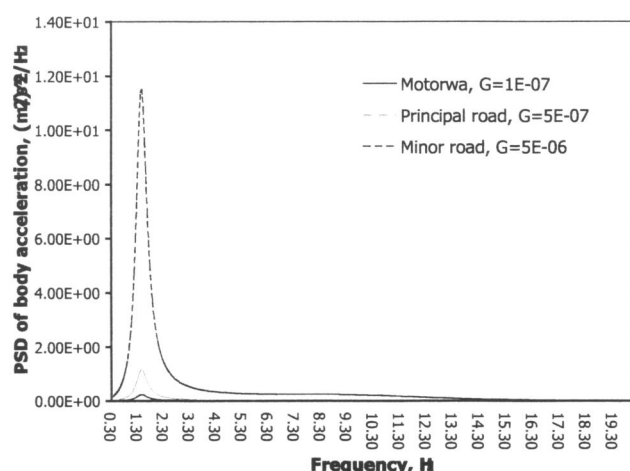
	$c=0.8 c_n$	$c = c_n$	$c=2 c_n$
$\dot{z}_s$	0.8136	0.7682	0.7555
<i>sws</i>	0.0105	0.0093	0.0065
<i>dtl</i>	6927.6	6330	5500



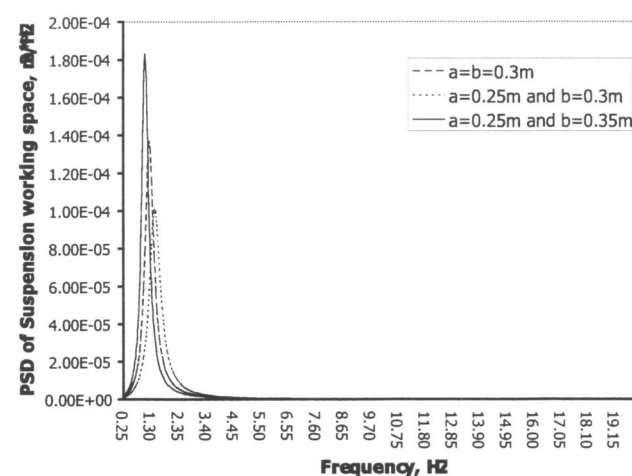
a- Vertical body acceleration



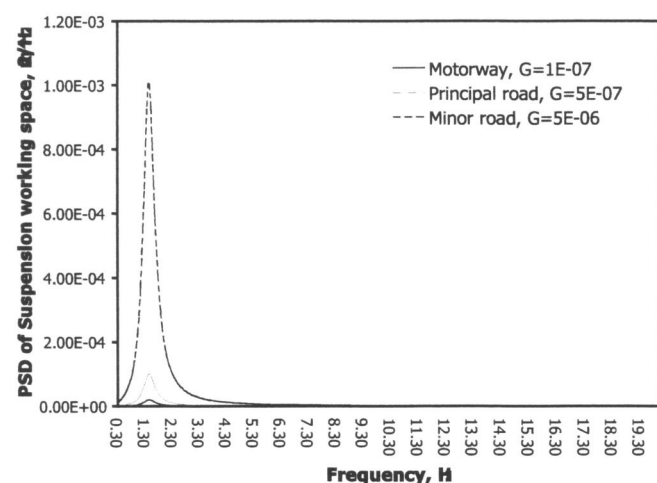
a- Vertical body acceleration



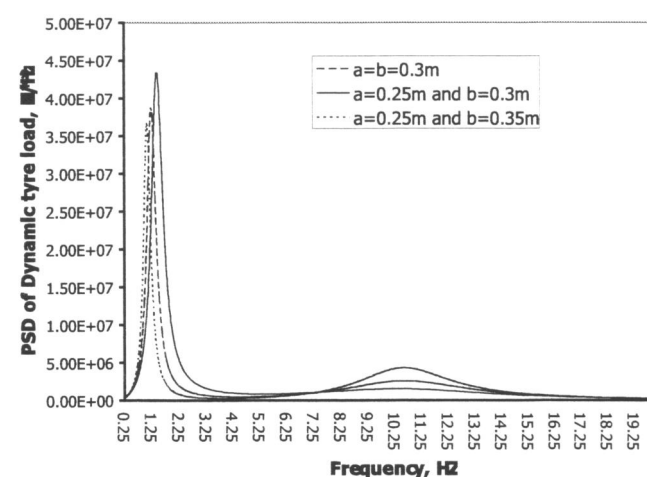
b- Suspension working space



b- Suspension working space



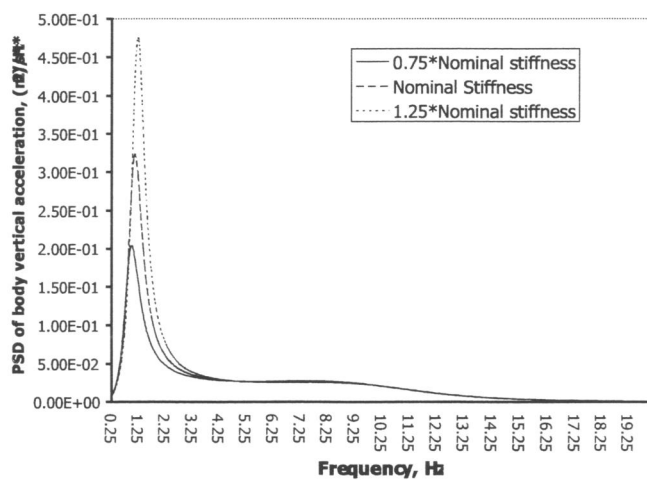
c- Dynamic tire load



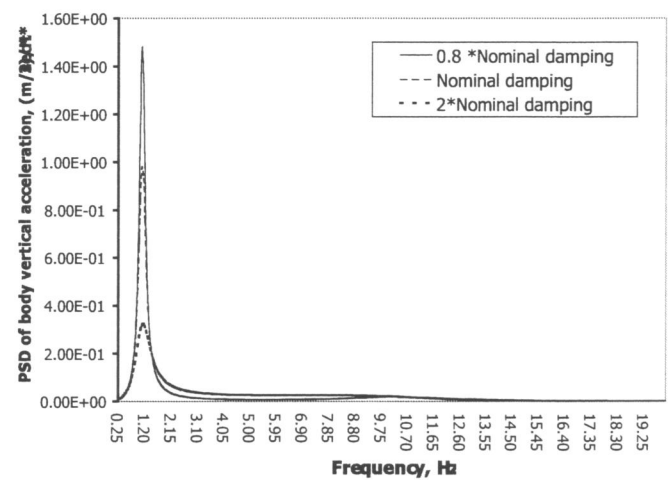
c- Dynamic tire load

Fig. 5 Variation of response variables of conventional single wheel station and the new suspension design with leverage ratio of $a/b=1$ ($v=20$ m/sec.).

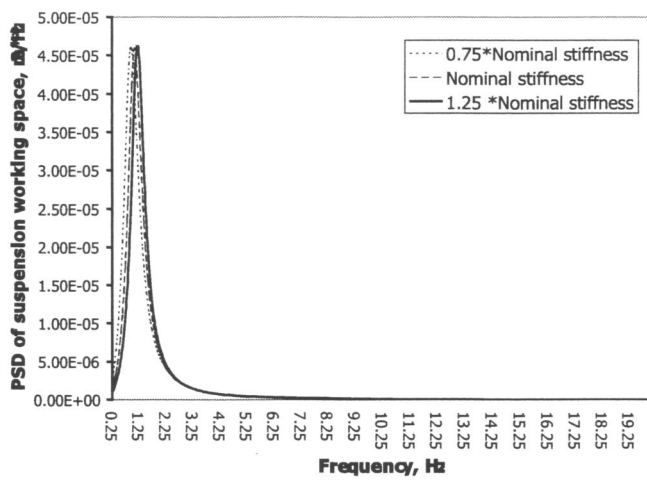
Fig. 6 Effect of leverage ratio on the response variables of the new single wheel model ($G=5 \times 10^{-7}$, $v=20$ m/sec.).



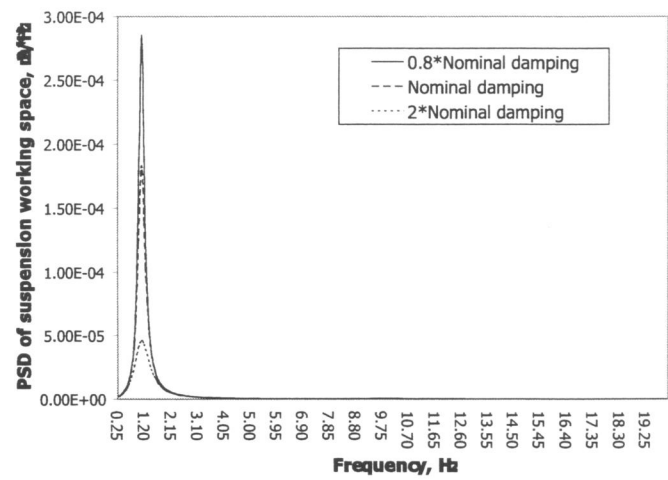
a- Vertical body acceleration



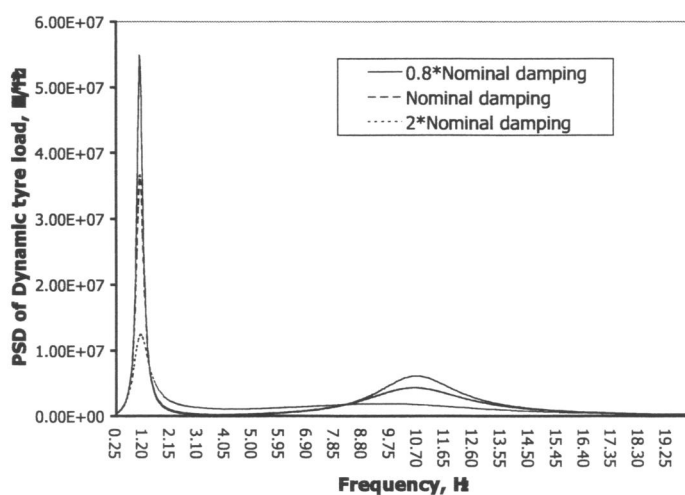
a- Vertical body acceleration



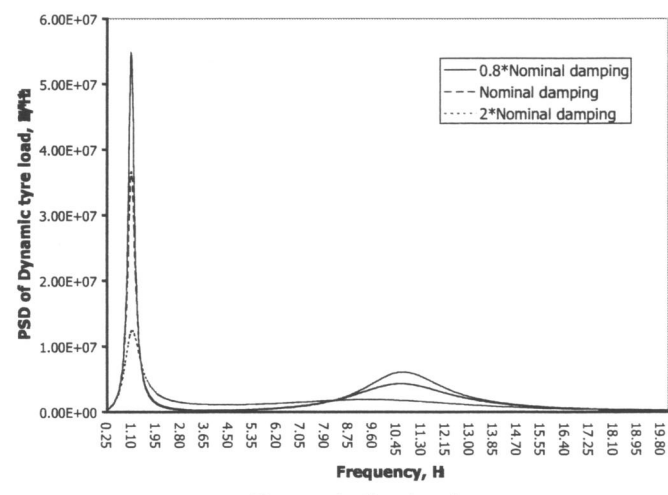
b- Suspension working space



b- Suspension working space



c- Dynamic tyre load



c- Dynamic tyre load

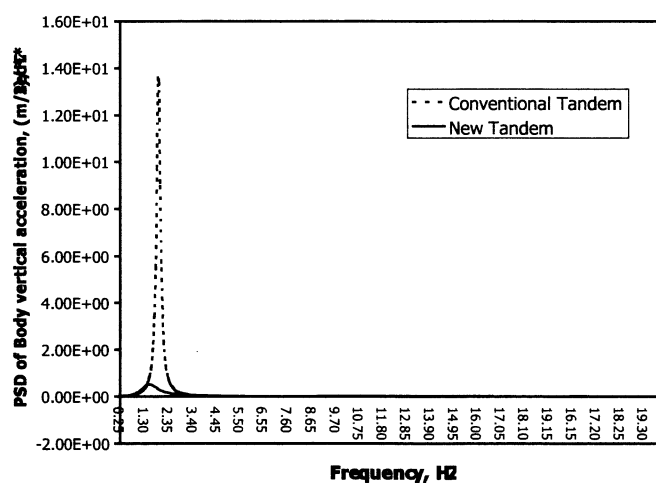
Fig. 7 Effect of suspension stiffness on the response variables of the new single wheel model ($G=5 \times 10^{-7}$, $v=20$ m/sec.).

Fig. 8 Effect of suspension damping on the response variables of the new single wheel model ($G=5 \times 10^{-7}$, $v=20$ m/sec.).

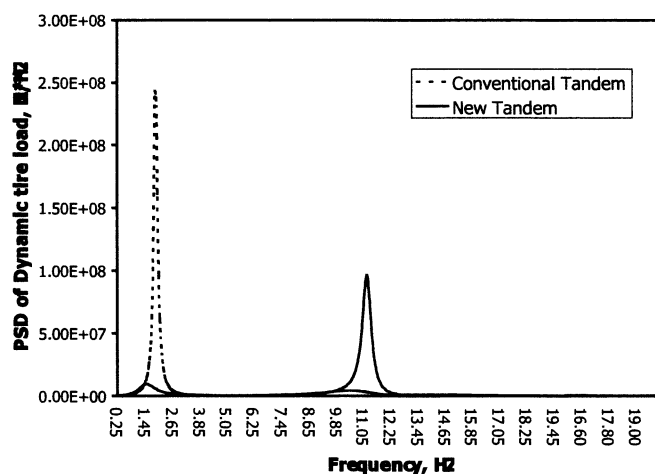
NEW TANDEM SUSPENSION SYSTEM

In this model there is an interaction between the two opposite bell crank mechanisms that led to a need for another round of trials to calculate the optimum suspension parameters.

After a large number of trials, it is found that the best leverage ratio for this design is $a/b = 0.25/0.5$. The horizontal springs and dampers are also selected carefully to reduce the dynamic tire load and the body vertical acceleration. A value of the spring stiffness of 0.5 of the equivalent nominal value and a value of the damping coefficient of 1.75 times its nominal value are found to give minimum system responses, Fig. 9 a & b. The eigenvalues (natural frequencies) of the coordinates z_s , θ_{c1} and θ_{c2} are 1.4, 11.5 and 11.7 Hz respectively. Good agreement has been found between these values and the observed values in the power spectral density curves.



a- Body acceleration



b- Dynamic tire load

Fig. 9 Vehicle response considering conventional and new tandem suspension systems ($G=5 \times 10^{-7}$, $v=20$ m/sec.).

CONCLUSIONS AND SUGGESTIONS FOR FURTHER WORK

From comparisons between the configurations of the response variables (Power spectral densities, PSD, and root mean square values, RMS) for both conventional and new suspension systems, the following conclusion can be drawn:

NEW SINGLE WHEEL STATION SUSPENSION

The design of the new single wheel station system gives the following advantages over the conventional one from the following points of view:

- Simple to control the system vibration levels by the variation of the leverage ratio of the bell crank (a/b).
- Reducing the levels of the power spectral density curves and hence reducing the root mean square values of the body vertical acceleration, suspension working space and dynamic tire load.
- For further work, the suspension mechanism (bell crank) can be used directly to design a new liftable axle/suspension system.

NEW TANDEM SUSPENSION SYSTEM

Based on the new system configuration and suspension parameters, the following conclusions can be stated:

- Significant improvements are achieved in the body acceleration, suspension working space and dynamic tire loads (58%, 33%, 5% respectively) over those of the conventional tandem suspension system.
- Reduction in the system degrees of freedom that makes the control of the vibration easy and reduces the system complexity.
- The frequency of the peak value of the dynamic tire load is moved from 2 Hz (conventional tandem suspension) to 11 Hz (new tandem suspension). This reduces the coupling between the vibration levels of both body and wheel/axle assemblies. Therefore, for further work, the application of the slow active suspension can be employed for such new tandem suspension system for a limited bandwidth of frequency.

REFERENCES

1. Inoh, Takeshi and Aisaka, Masahru (1973) 'Tuning technique for controlling heavy-duty truck shak-vertical, torsional, and lateral' SAE paper No. 730650.

2. Baum, J. H., Bennett, J. A. and Carne, T. G. (1977) 'Truck ride improvement using analytical and optimization methods' SAE paper No. 770609.
3. Foster, Albert W. (1978) 'A heavy truck cab suspension for improved ride' SAE paper No. 780408.
4. Dokainish, M. A. and El-Madany, M. M. (1980) 'Random response of tractor-semitrailer system' Vehicle System Dynamics, Vol. 9, pp. 87-112.
5. El-Madany, M. M. (1987) 'An analytical investigation of isolation systems for cab ride' Computers & Structures, Vol. 27, No. 5, pp. 679-688.
6. Bruce, D. Van Deusen, (1971) 'Truck suspension optimization' SAE paper No. 710222.
7. El-Madany, M. M. (1988) 'Design and optimization of truck suspensions using covariance analysis' Computers & Structures, Vol. 28, No. 2, pp. 241-246.
8. Pintado, P. and Benitez, F. G. (1990) 'Optimization of vehicle suspension I: Time domain' Vehicle System Dynamics, Vol. 19, pp. 273-288.
9. Del Castillo, J. M., Pintado, P. and Benitez, F. G. (1990) 'Optimization of vehicle suspension II: Frequency domain' Vehicle System Dynamics, Vol. 19, pp. 331-352.
10. Rakheja, S. Ahmed, A. K. W. and Yang, X. (1999) 'Optimal suspension damping for improved driver- and road friendliness of urban buses' SAE paper No. 1999-01-3728, SP-1486, Roll stability dynamics, Ride, and Handling.
11. El-madany, M. M. and Samaha, M. E. (1992) 'On the optimum ride control of a stochastic model of a tractor-semitrailer vehicle' Journal of Sound and Vibration, Vol. 156, No. 2, pp. 269-281.
12. Wilkinson, P. A. and Crolla, D. A. (1992) 'Development of sub-optimally controlled suspensions for articulated truck to reduce vibration and road wear' XXIV FISITA Congress, London, pp. 239-245.
13. Cebon, D., Besinger, F. H. and Cole, D. J. (1996) 'Control strategies for semi-active lorry suspensions' Proc Instn Mech. Engrs, Part D: Journal of Automobile Engineering, pp. 161-178.
14. Ibrahim, I. M. (1999) 'Finite element multibody system control of tractor semi-trailers with active suspension and controller time delay' SAE paper No. 1999-01-0726, SP-1438, Steering and Suspension Technology Symposium.
15. Crolla, D. A., Firth, G. and Horton, D. (1993) 'An Introduction to vehicle dynamics' Mechanical Engineering Dept., Leeds University, UK.
16. Ibrahim, I. M., Crolla, D. A. and Barton, D. C. (1995) 'Active suspension for trucks incorporating frame flexibility effects' Heavy Vehicle Systems, Special Series of the Int. J. of Vehicle Design, Vol. 2, No. 1, pp. 1-17.
17. Potter, T. E. C., Cebon, D., Collop, A. C. and Cole, D. J. (1996) 'Road-damaging potential of measured tyre forces in mixed traffic' Proc Instn Mech Engrs,

Part D: Journal of Automobile Engineering, pp. 215-225.

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LIST OF SYMBOLS

The majority of the notations are defined within the text next to the equations in which they are first used. Additional notations are listed here:

a : vertical arm distance of the bell crank, m
 b : horizontal arm distance of the bell crank, m
 z_s : vertical displacement of the vehicle body,
 z_u : vertical displacement of the wheel/axle assembly,
 θ_u : angular displacement of the suspension arm of the conventional tandem suspension system,
 θ_b : angular displacement of the balancing arm of the conventional tandem suspension system,
 θ_c : angular displacement of the bell crank,
 k_s : suspension stiffness coefficient,
 c_s : suspension damping coefficient,
 k_n : nominal value of suspension stiffness coefficient,
 c_n : nominal value of suspension damping coefficient,
 k_t : tire stiffness coefficient,
 c_t : tire damping coefficient,
 v : vehicle velocity,
 G : is the roughness coefficient,
 f : frequency,
 I_c : mass moment of inertia of the bell crank,
 $[M]$, $[C]$, $[K]$: mass, damping and stiffness matrices
 $\{z\}$ is the vector of the system generalized coordinates,
 $\{z_o\}$ is the vector of the road excitation signals,
 $[C_{EX}]$, $[K_{EX}]$: are excitation distribution matrices of the road excitation signals,
 PSD : power spectral density,
 RMS : root mean square value,
 sws : suspension working space,
 d_{tl} : dynamic tire load.